Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28 September 2026
Team ID	SWTID-2026-2169
Project Name	Import Data Using Transform Map
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through Linked IN
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Transform Map Configuration	Create a Transform Map from the Import Set table to the target table Map source fields to target fields (auto map and manual map) Map reference fields such as department or user
FR-4	Duplicate Handling	Set a coalesce field (for example Email or Employee ID) Choose the action: create new records or update existing records
FR-6	Data Validation and	Clean and format data using transform scripts (trim spaces, convert date formats) Ignore or flag invalid rows

	Transformation	
FR-7	Error Monitoring and Reporting	Check the Transform History and Import Set Row status View error logs and rerun failed rows Send email notification when the import is complete

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Import Set and Transform Map screens are simple and easy to use. Admins can upload an Excel/CSV file, map source fields to target table fields, and run the transform with minimal training. Field mapping is shown clearly, and error messages are easy to understand.
NFR-2	**Security**	Only users with authorized roles (e.g., admin, import admin) can create data sources, transform maps, and run imports. ACLs protect the staging (import set) and target tables. Data is transferred over secure HTTPS, and all import activity is logged for audit.
NFR-3	**Reliability**	The transform map consistently maps and loads data into the target table without loss or duplication. Coalesce fields prevent duplicate records, and transform errors and skipped rows are logged in the Import Set Row and Transform History for review and re-run.
NFR-4	**Performance**	Imports process large record volumes within an acceptable time. Field maps and transform scripts are kept lightweight, and imports can be scheduled during off-peak hours to avoid affecting the instance's performance.
		Data import is available whenever needed through manual runs or scheduled imports. Because it runs on the Service

NFR-5	Availability	Now cloud platform, it benefits from the platform's high uptime, and failed imports can be re-run without downtime.
NFR-6	Scalability	The solution handles growth in data volume and number of data sources. New transform maps, field mappings, and import sets can be added easily, and the same approach works for multiple target tables (e.g., Incident, User, Group) without redesign.